

Ex:No:02	BOOK BANK REGISTRATION SYSTEM
Date:	

AIM

To draw the UML diagrams such as Use Case, Class, Activity, Sequence, Collaboration, State Chart, Component, Deployment and Package diagrams for the **Book Bank Registration System**.

SOFTWARE REQUIREMENTS SPECIFICATION

1.0 Hardware Requirements

- Intel Pentium Processor i3/i5
- 4 GB RAM
- 500 GB Hard Disk

1.1 Software Requirements

- Umbrello UML Modeller
- Windows/Linux Operating System

Note: Your original record mentions **Rational Rose / Argo UML**. Since you're actually creating the diagrams in **Umbrello**, writing Umbrello here is appropriate for your practical work. The source lists Rational Rose/Argo UML.

1.2 PROBLEM ANALYSIS AND PROJECT PLAN

The system is designed to simplify the process of managing books in the book bank. The student details are provided to the administrator and verified by the computer. After verification, the required books are checked for availability and issued to the student.

1.3 PROJECT DESCRIPTION

The **Book Bank Registration System** is designed to verify student details and manage the issue of books. The administrator provides the student details to the computer, which verifies the details and provides approval. The required books are then issued to the student from the book bank.

1.4 REFERENCE

IEEE Software Requirement Specification format.

USE CASE DIAGRAM

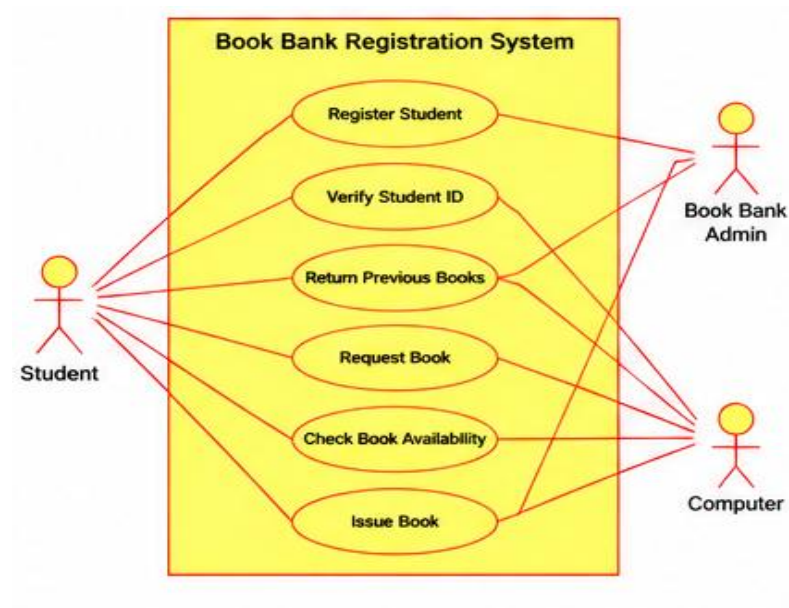
Actors:

- Student
- Book Bank Admin
- Computer

Use Cases:

- Student Details
- Register
- Verify Student ID
- Return Previous Books
- Request Book
- Check Book Availability
- Issue New Books

These are the actors and use cases specified in your record.



CLASS DIAGRAM

The Class Diagram

consists of the following classes, attributes and operations.

Class	Attributes	Operations
Computer	Student Record, Book List	Enter Issue(), Check Availability()
Student	Student Details	Request for Books(), Register(), Return Previous Books()
Admin	Student Details, Book List	Verify Student ID(), Check Book Availability(), Issue Books(), Order for New Author(), Maintain Student Details()

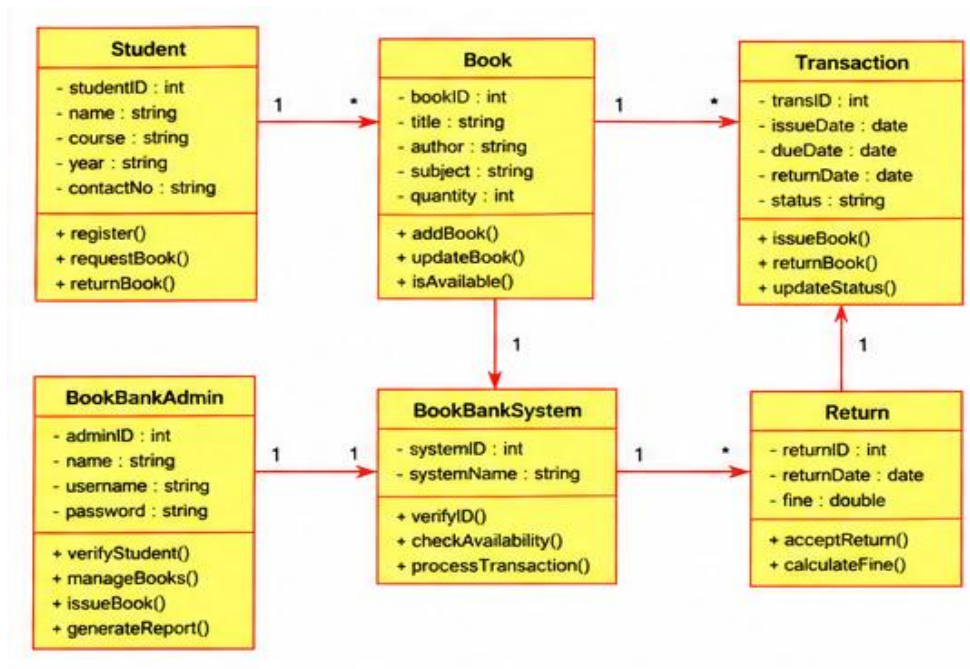
ACTIVITY DIAGRAM

Activities:

1. Verify ID
2. Return Books
3. Request for Books
4. Check for Availability
5. Enter Book Issue Details
6. Issue Books

Decision:

Check whether the required book is available or not.



SEQUENCE DIAGRAM

The Sequence Diagram consists of objects, messages and return messages.

Objects:

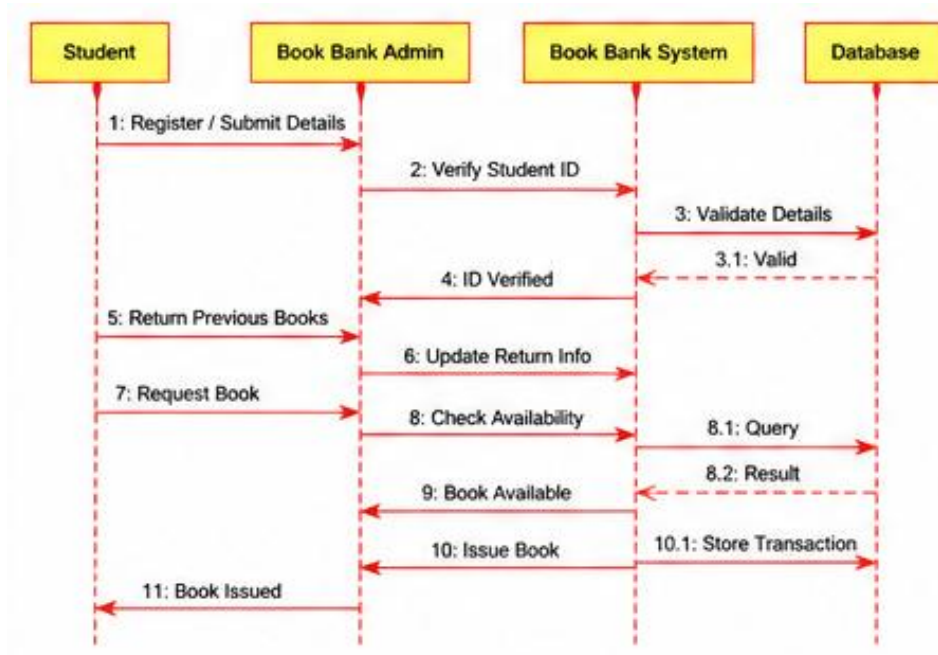
- Student
- Admin
- Computer

Sequence:

1. Submit ID
2. Verify ID
3. Valid ID
4. Return Books
5. Request Books
6. Check Book Availability
7. Book Availability Status

8. Enter Book Issue Details

9. Issue New Books



Step Description

- 1 Student submits ID
- 2 Admin verifies ID
- 3 Student requests books
- 4 Computer checks availability
- 5 Books are issued

COLLABORATION DIAGRAM

The Collaboration Diagram represents the communication between the objects and actors of the Book Bank Registration System. It is obtained from the sequence diagram.

Objects:

- Student

- Admin
- Computer

Messages:

1. Submit ID
2. Verify ID
3. Valid ID
4. Return Books
5. Request Books
6. Check Book Availability
7. Book Availability Status
8. Enter Book Issues
9. Issue New Books

STATE CHART DIAGRAM

The State Chart Diagram represents the different states involved in the book issuing process.

States:

Start → Give Details → Register → Select Books → Check Availability → Issue Book → End

Paste your State Chart Diagram here.

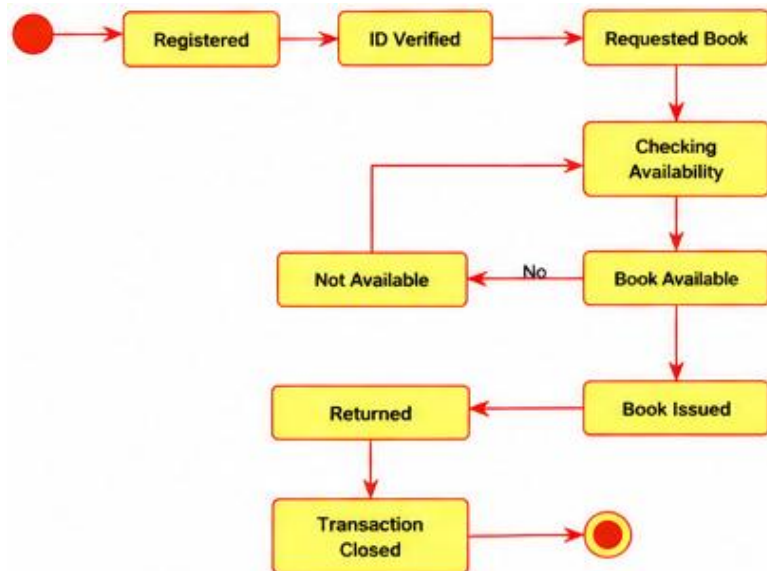
COMPONENT DIAGRAM

The Component Diagram represents the structural relationships between the components of the Book Bank Registration System. Dependencies are represented using communication associations.

Components:

- Book Bank
- Member Details

- Issue for 1st Year
- Issue for 2nd Year
- Issue for 3rd Year
- Issue for 4th Year



DEPLOYMENT DIAGRAM

The Deployment Diagram represents the physical deployment of artifacts on deployment targets.

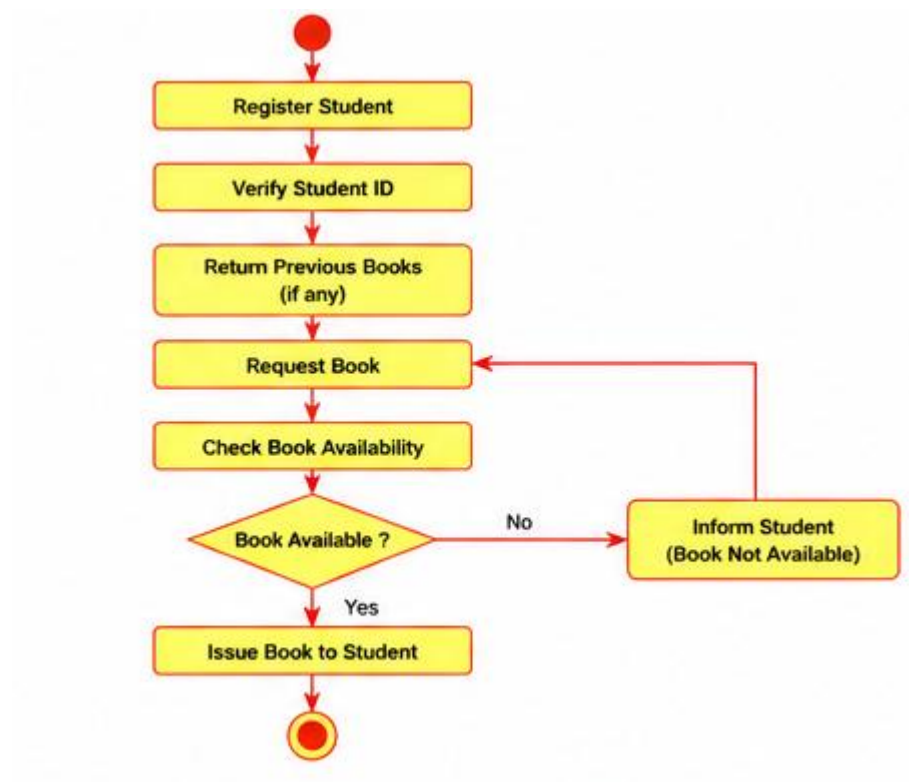
Nodes/Elements:

- Book Bank
- Member Details
- Issue for 1st Year
- Issue for 2nd Year
- Issue for 3rd Year
- Issue for 4th Year

ACTIVITY DIAGRAM

The Package Diagram represents the dependencies between the packages that make up the system. It contains three layers:

1. **User Interface Layer**
2. **Domain Layer**
3. **Technical Services Layer**



PROGRAM CODING

ADMIN

```
Public class admin
```

```
{
```

```
    Public integer student details;
```

```
    Public integer book list;
```

```
    Public void verify stud id()
```

```
{  
}
```

Public void check for availability()

```
{  
}
```

Public void issue books()

```
{  
}
```

Public void order for new author()

```
{  
}
```

Public void maintains stud details()

```
{  
}
```

```
}
```

This follows the coding section in your record.

STUDENT

Public class stud

```
{
```

Public integer studdetails;

```
Public void request for books()
```

```
{
```

```
}
```

```
Public void register()
```

```
{
```

```
}
```

```
}
```

COMPUTER

```
Public class computer
```

```
{
```

```
Public integer stud record;
```

```
Public integer booklist;
```

```
Public void maintain stud rec()
```

```
{
```

```
}
```

```
Public void enter issue()
```

```
{
```

```
}
```

```
Public void order new author()
```

```
{
```

```
}
```

```
Public void check availability()
```

```
{
```

```
}
```

```
}
```

The Student and Computer coding are from the corresponding sections of your record.

RESULT

Thus, the Use Case, Activity, Sequence, Collaboration, Class, State Chart, Component, Deployment and Package diagrams for the Book Bank Registration System were successfully designed, executed and verified.